Technische Universität München

Leistungsnachweis
Grade Report

Familienname/ Family Name:
Sakalyan

Vorname(n)/ First Name(s):
Kristiyan

Geburtsdatum/ Date of Birth:
21. Juli 1999
21 July 1999

Geburtsort/ Place of Birth:
Dulovo

Matrikelnummer/ Student ID Number:
03714584

Angestrebter Abschluss/ Degree in progress:
Master of Science (M.Sc.)

Studiengang/ Degree Program:
Data Engineering and Analytics
Data Engineering and Analytics

Datum/ Date:
18. September 2025
18 September 2025

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This document was created using an electronic data processing system and is therefore valid without a signature. Where indicated, the document has been signed electronically with an advanced signature.

Aktuelle Gesamtcredits Current Total Credits	**125**
Zwischennote aus den in die Notenberechnung eingegangenen Modulen Provisional Grade according to Grade-Relevant Modules	**1,2**
Dies ist kein Abschlussdokument. This is not an official graduation document.	

Modul-ID Module ID	**Bezeichnung** Title	**Note** Grade	**Credits** Credits
Master's Thesis Master's Thesis			
IN2327	**Master's Thesis** Master's Thesis	**1,0**	**30**
	Thema: Modeling Cellular Trajectories on Time-Resolved Spatial Data Using Optimal Transport and Flow Matching Die Thesis wurde in englischer Sprache verfasst. Topic: Modeling Cellular Trajectories on Time-Resolved Spatial Data Using Optimal Transport and Flow Matching The thesis was written in English.		
	Master's Thesis Master's Thesis	1,0	

Modul-ID Module ID	Bezeichnung Title		Note Grade	Credits Credits
Pflichtmodule Data Engineering and Analytics Required Modules Data Engineering and Analytics				
MA4800	**Foundations of Data Analysis** Foundations of Data Analysis		**3,3**	**8**
	Foundations of Data Analysis Foundations of Data Analysis	3,3		
IN2326	**Foundations in Data Engineering** Foundations in Data Engineering		**1,7**	**8**
	Foundations in Data Engineering Foundations in Data Engineering	1,7		
Master-Praktikum Advanced Practical Course				
IN2106	**Master-Praktikum** Advanced Practical Course		**1,0**	**10**
	Master Praktikum - Machine Learning for Natural Language Processing Applications Master Lab Course - Machine Learning for Natural Language Processing Applications	1,0		
Master-Seminar Advanced Seminar Course				
IN2107	**Master-Seminar** Advanced Seminar Course		**1,0**	**5**
	Seminar - Selected Topics in Machine Learning Research Seminar - Selected Topics in Machine Learning Research	1,0		
Wahlmodule Elective Modules				
Data Engineering Data Engineering				
IN2219	**Query Optimization** Query Optimization		**1,0**	**6**
	Anfrageoptimierung Query Optimization	1,0		
Data Analytics Data Analytics				
IN2064	**Machine Learning** Machine Learning		**1,7**	**8**
	Maschinelles Lernen Machine Learning	1,7		

Modul-ID Module ID	Bezeichnung Title		Note Grade	Credits Credits
Data Analysis Data Analysis				
CIT5130001	**Applied Statistics and Data Analysis (TUM School of Computation, Information and Technology [CIT] and TUM School of Life Sciences [SoLS])** Applied Statistics and Data Analysis (TUM School of Computation, Information and Technology [CIT] and TUM School of Life Sciences [SoLS])		**1,0**	**5**
	Applied Statistics and Data Analysis Applied Statistics and Data Analysis	1,0		
Advanced Topics in Data Engineering Advanced Topics in Data Engineering				
IN2328	**Anwendungsprojekt** Application Project		**1,0**	**10**
	TUM Data Innovation Lab TUM Data Innovation Lab	1,0		
CIT3230003	**Data Structure Engineering** Data Structure Engineering		**1,3**	**5**
	Data Structure Engineering Data Structure Engineering	1,3		
Special Topics in Data Analytics Special Topics in Data Analytics				
CIT4230003	**Advanced Machine Learning: Deep Generative Models** Advanced Machine Learning: Deep Generative Models		**1,0**	**3**
	Advanced Machine Learning: Deep Generative Models Advanced Machine Learning: Deep Generative Models	1,0		
CIT4230002	**Advanced Natural Language Processing** Advanced Natural Language Processing		**1,0**	**5**
	Advanced Natural Language Processing Advanced Natural Language Processing	1,0		
IN2323	**Machine Learning for Graphs and Sequential Data** Machine Learning for Graphs and Sequential Data		**1,0**	**5**
	Machine Learning for Graphs and Sequential Data Machine Learning for Graphs and Sequential Data	1,0		
Wahlmodulkatalog Informatik Elective Modules Informatics				
IN2309	**Advanced Topics of Software Engineering** Advanced Topics of Software Engineering		**1,0**	**8**
	Advanced Topics of Software Engineering Advanced Topics of Software Engineering	1,0		

Modul-ID Module ID	Bezeichnung Title		Note Grade	Credits Credits	
Überfachliche Grundlagen Support Electives					
SZ1201	**Spanisch A1** Spanish A1		**1,3**	**3**	
	Spanisch A1 Spanish A1	1,3			
POL20401	**Political Data Science - Project (Bachelor)** Political Data Science - Project (Bachelor)		**1,0**	**6**	
	Political Data Science - Project Political Data Science - Project	1,0			

Erläuterungen/Explanations:

Notenskala:1,0-1,5 sehr gut, 1,6-2,5 gut, 2,6-3,5 befriedigend, 3,6-4,0 ausreichend, 4,1-5,0 nicht ausreichend
Grades:1,0-1,5 very good, 1,6-2,5 good, 2,6-3,5 satisfactory, 3,6-4,0 sufficient, 4,1-5,0 fail

Bewertung von Studienleistungen: BE = bestanden NB = nicht bestanden
Performance Key: BE = pass NB = fail

Credits: Gemäß dem European Credit Transfer System (ECTS) Maßeinheit für die Arbeitsbelastung eines Studierenden; ein Credit entspricht der Arbeitszeit von 30 Stunden.
Credits: a unit of measure within the European Credit Transfer System (ECTS) representing student workload. A credit is equal to 30 hours of work.

Module ohne zugeordnete Note und Credits sind noch nicht vollständig bestanden. Sind Teilnoten mit dem Wert "nicht ausreichend" (4,1-5,0) angeben, so gilt die Ausgleichsregelung: Das Modul ist auch dann bestanden, wenn nicht alle Modulteilprüfungen bestanden sind, sofern die Modulnote 4,0 oder besser ist. Für die Gewichtung der Modulteilprüfungen, die Berechnung der Gesamtnote sowie weitere Informationen siehe die Fachprüfungs- und Studienordnung für diesen Studiengang in der gültigen Fassung sowie das Modulhandbuch.
Where grades and credits have not been assigned to modules, the student has not yet successfully completed all required module components. Component grades designated as "fail" (4,1-5,0) are subject to the compensation rule: The module is considered passed even if the student does not pass all module examination components provided that the student's grade for the module is 4,0 or better. For further information and details on the weighting of module examination components, as well as the calculation of the overall grade, please refer to the current Academic and Examination Regulations of the relevant degree program.

*) = anerkannt **) = enthält anerkannte Leistungen
*) = accredited **) = contains accredited exams

Technische Universität München

Leistungsnachweis: Zusatzleistungen
Grade Report: Additional Credits

Familienname/ Family Name:
Sakalyan

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Kristiyan

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21. Juli 1999
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Modul-ID Module ID	**Bezeichnung** Title	**Note** Grade	**Credits** Credits
Additional Credits Additional Credits			
	Business Analytics and Machine Learning Business Analytics and Machine Learning	1,8	5 *)
	Cloud-Based Data Processing Cloud-Based Data Processing	1,3	5
	Introduction to Deep Learning Introduction to Deep Learning	1,3	6
	Advanced Deep Learning for Computer Vision: Visual Computing Advanced Deep Learning for Computer Vision: Visual Computing	2,3	8
	Natural Language Processing Natural Language Processing	1,3	6

Erläuterungen/Explanations:

Notenskala:1,0-1,5 sehr gut, 1,6-2,5 gut, 2,6-3,5 befriedigend, 3,6-4,0 ausreichend, 4,1-5,0 nicht ausreichend
Grades:1,0-1,5 very good, 1,6-2,5 good, 2,6-3,5 satisfactory, 3,6-4,0 sufficient, 4,1-5,0 fail

Bewertung von Studienleistungen: BE = bestanden NB = nicht bestanden
Performance Key: BE = pass NB = fail

Credits: Gemäß dem European Credit Transfer System (ECTS) Maßeinheit für die Arbeitsbelastung eines Studierenden; ein Credit entspricht der Arbeitszeit von 30 Stunden.
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Alle in dieser Anlage aufgeführten Ergebnisse gehen über die für das Bestehen des Studiengangs erforderlichen Leistungen hinaus. Die erzielten Noten und Credits fließen nicht in das Gesamtergebnis des Studiengangs ein.
The modules and courses listed on this document are not required for the successful completion of the degree program. As such, the grades and credits earned for these modules are not included in the calculation of the student's overall grade and credit total.